Identification: excite, estimate, then challenge the model

02 / Source-reported parameter sets and separate synthetic identification experiments

Source parameter register: four distinct FOPDT models

Case	K	τ [s]	θ [s]
TH37	8.7135	326.18	13.677
TH34	7.7473	177.95	13.064
Mean measured temperature	7.624	151.6	8.78
Mean estimated parameters	8.028	186.5	12.5

The mean of sensor responses and the mean of fitted parameters need not have the same dynamics. TH22 and TH40 are not independently identified here.

Illustrative excitation u_k: record sample times, ambient offset, input units and delay convention

Two explicitly different estimators

Notebook 01 fits an analytical FOPDT step response to synthetic data. Notebook 05 uses the restricted known-integer-delay ARX model below. Neither experiment recreates the historical MATLAB fit or recovers missing original input/temperature records.

Freeze fitted coefficients

ARX(1,1) with a known integer delay m

$z_k = y_k - T_a, \quad z_k = a z_{k-1} + b u_{k-1-m} + \varepsilon_k.$

$\Phi_k = [z_{k-1} \quad u_{k-1-m}], \quad \hat{\beta} = \arg\min_{(a,b)} \|\Phi\beta - z\|_2^2.$

$\hat{\tau} = -\Delta t / \ln \hat{a}, \quad \hat{K} = \hat{b}/(1-\hat{a}), \quad \hat{\theta} = m\Delta t.$

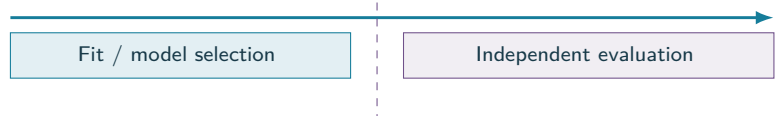
Mapping requires $0 < \hat{a} < 1$, $\hat{b} > 0$, full-rank excitation and compatible units. The integer-delay formulation does not reproduce a fractional delay exactly.

Recommended extension: independent evaluation

One step: predict with the measured previous output. A low residual can hide dynamic mismatch.

Free run: feed earlier predictions back into the model. Check accumulated bias, delayed onset and long-term gain.

Not executed by notebooks 01/05: their fitted residual RMSE is in-sample. Freeze model choices before an independent test.



Recommended split; not executed by notebooks 01/05

Unresolved historical inputs

Original synchronized heater drive, raw sensor readings and sufficiently documented sampling/validation protocol are not supplied. The diagrams and synthetic fits do not establish the thesis's reported fit percentages.

The command trace is a conceptual excitation sketch, not a dataset. Parameter values follow the repository's source register; estimator equations follow identification.py and the notebook-specific protocol. Original explanatory drawing for ThermoFuzzy-PID. Source traceability: Cordeiro (UFC, 2021), documented repository equations and explicit reconstruction choices.